

Name:

Enrolment No:

UPES

End Semester Examination (Even Semester), May 2026

Course: B.Tech. (SoCS)

Program: Advanced Engineering Mathematics-2

Course Code: MATH1065

Semester : II

Time : 03 hrs.

Max. Marks: 100

Instructions: Attempt all the Questions. Questions 6 & 10 include internal choices. Calculators are allowed.

SECTION A
(5Qx4M=20Marks)

S. No.		Marks	CO
Q. 1	Apply the method of separation of variables to form two ordinary differential equations corresponding to the partial differential equation $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0.$	04	CO5
Q. 2	Find the zeros and poles of $f(z) = \left(\frac{z+1}{z^2+1}\right)^2$. Also, determine the order of each pole.	04	CO2
Q. 3	Represent $f(t) = \begin{cases} \sin 2t, & 2\pi < t < 4\pi \\ 0, & \text{otherwise} \end{cases}$, in terms of unit step function.	04	CO4
Q. 4	Expand the function $f(z) = \frac{1}{(z+4)}$ in a Taylor's series expansion valid for the region $ z < 4$.	04	CO2
Q. 5	Convert the following Linear programming problem into the standard form $\min z = 5x_1 + 2x_2 - 3x_3$ subject to the constraints $x_1 + 2x_2 - 5x_3 \leq 5$ $x_2 - x_3 \geq -4$ $5x_1 + 2x_3 \leq 10$ where $x_1, x_2, x_3 \geq 0$.	04	CO1

SECTION B
(4Qx10M= 40 Marks)

Q. 6	The values of a quantity Q at $t = 3, 4, 5, 6$ are given by <table><tr><td>t</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$Q(t)$</td><td>16</td><td>20</td><td>24</td><td>38</td></tr></table> Estimate the value of Q at $t = 3.2$ using appropriate Newton's difference formula. OR Find the root of the equation $\cos x = xe^x$ using Newton-Raphson method with initial approximation $x_0 = 1$ correct up to 3 decimal places.	t	3	4	5	6	$Q(t)$	16	20	24	38	10	CO1
t	3	4	5	6									
$Q(t)$	16	20	24	38									
Q. 7	Analyze the differential equation $x^2(x-1)^2y'' + (x-1)y' + y = 0$ to determine all singular points and classify them as regular or irregular.	10	CO3										

Q. 8	Evaluate $\oint_C \frac{3z e^z}{z^2 - 1} dz$ around the circle $C: z = 2$ in counterclockwise direction.	10	CO2
Q. 9	Determine the analytic function whose real part is $e^{2x}(x \cos 2y - y \sin 2y).$	10	CO2
SECTION-C (2Qx20M=40 Marks)			
Q. 10	a) The equation of motion of a vehicle is given by differential equation $\frac{d^2 y}{dt^2} + y = t^2 + 2t.$ It is known that the values of y and $\frac{dy}{dt}$ at $t = 0$ are 4 and -2 respectively. Apply Laplace transformations to solve this differential equation. b) Find Inverse Laplace transform of the function $F(s) = \frac{4s + 15}{16s^2 - 25}.$ OR Derive the Fourier series of $f(x) = \begin{cases} x - \pi, & -\pi < x < 0 \\ \pi - x, & 0 < x < \pi \end{cases}$ in the interval $(-\pi, \pi)$ and hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$	20	CO4
Q. 11	A string is stretched and fastened at two points l distance apart. Motion is started in the string by displacing it the form $y = a \sin \frac{\pi x}{l}$ from which it is released at a time $t = 0$. Assume that the motion of the string is governed by the wave equation $y_{tt} = c^2 y_{xx}$. State the general solution $y(x, t)$ of the problem (without proof). Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin \left(\frac{\pi x}{l} \right) \cos \left(\frac{\pi c t}{l} \right).$	20	CO5